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**The Development of a Medical Robot for Automated
Ultrasound Scan**

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Introduction

A robot is a machine capable of carrying out a complicated set of activities automatically, particularly one that can be programmed by a computer. A robot can be steered by an external control device or by an integrated control device. Although some robots are meant to resemble humans, most are task-oriented devices with a focus on bare utility rather than expressive aesthetics.

Humanoids like Honda's Advanced Step in Innovative Mobility (ASIMO) and TOSY's TOSY Ping Pong Playing Robot (TOPIO) to industrial robots, medical operating robots, patient assist robots, dog therapy robots, collectively programmed swarm robots, UAV drones like General Atomics' MQ-1 Predator, and even microscopic nano robots are examples of autonomous or semi-autonomous robots. A robot's feeling of intelligence or cognition can be conveyed by emulating a human look or automating motions. Autonomous items are projected to become more prevalent in the future, with home robots and autonomous vehicles being two chevaliers.

When it's come to medical application, there has been many innovations for surgery, examining and to comfort the patients. In this case, the review is based on robot manipulator used for ultrasound scan on patients abdominal.

Objectives

The main objective is to locate the patients and scan the patient's abdominal using ultrasound properties with the help of universal UR3 robot manipulator. The focus of this project is to apply the required force on skin of patients and with help of machine vision application, is to locate the position of the patients abdominal and scan for medical purpose. The process can be carried out by testing the working of manipulator with required force in simulation software like Abb robotstudio or any other resource available which is then followed by testing machine vision properties on real robot environment.

Literature Review

Medical Robots Example

Robotics and artificial intelligence (AI) have gradually made their way into the daily lives of the majority of the world's population. Medical robotics is one of the most successful instances of breakthrough technologies that have enabled the creation of new medical equipment. Medical robots have been used in a variety of settings to aid people with jobs that are monotonous, dangerous, or need a high level of accuracy or complicated skill. New medical robot applications are discovered and documented on a regular basis. The enormous advantages of medical robots have been confirmed in clinical diagnosis, surgical treatment, postoperative rehabilitation, home care, and other domains as medical care has evolved. Medical robots will undoubtedly become more essential in the next years. In some ways, medical robots have pervaded every element of human existence, and medical robot research and development continues to produce a wide range of medical robots with varying design aims, applications, structures, materials, and intelligence levels.

The eight important research themes in medical robotics during the last decade are identified and analysed such as robotic laparoscopy, non-laparoscopic robots for minimally invasive surgery, assistive wearable robots, therapeutic rehabilitation robot, capsule robots, magnetic actuation, soft robotics and continuum robotics in this Review article [1] (Dupont et al., 2021). These theme categories shown in figure 1 were discovered using search parameters that uncovered the decade's most referenced publications. They have concentrated only on a limited number of important articles in each theme area since their objective for this Review article is to give an easy approach for readers to rapidly appreciate some of the most exciting developments in medical robotics over the last decade. They hope that this article encourages researchers to be more entrepreneurial in order to bridge the gap between research and implementation. They also believe that Robotics is a discipline that looks ahead. The focus is on determining the next great challenges. However, in an applied subject like medical robotics, it's critical to plan for the future based on a thorough awareness of what the research community has recently done and where this work is in terms of clinical requirements and commercialization.

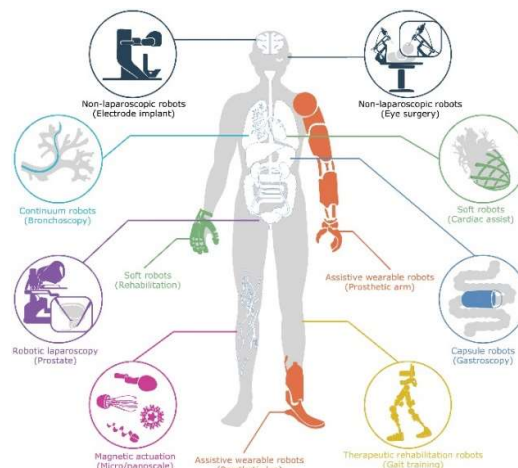


Figure 1

The design, implementation, and operating technique of an autonomous laser toning system known as MELON are suggested in this study [2] (Oh, Bae, Lim and Oh, 2016), (Manipulator for Effective Laser toning). Using intuitive differential inverse kinematics of the robot manipulator, the system can distinguish exact treatment sites from 3D point cloud data received with the camera, and it is feasible to emit the laser at the appropriate location and orientation frequently, precisely, and accurately. The MELON's feasibility test is carried out on a plaster cast of a woman's head, and they discover that the manipulator has enough workspace to cover the full face of the human inductively, and that the laser emission distribution is uniform throughout the face. From this proposal, the technique of laser toning can be applied into ultrasound scan.

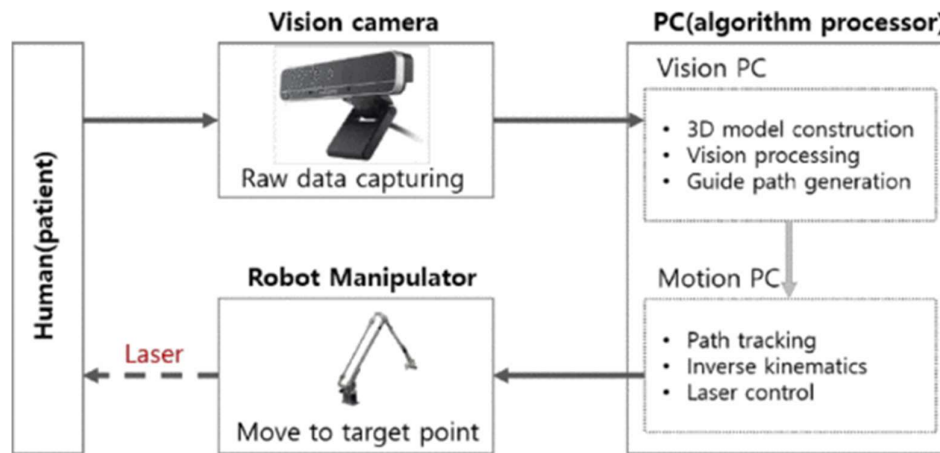


Figure 2

This paper [3] (Moustris, Hiridis, Deliparaschos and Konstantinidis, 2011) is based on **Background:** Autonomous control of surgical robotic platforms may improve accuracy, intelligent motions, tissue damage prevention, and other features. In surgery, autonomous robotic devices are mostly in the experimental stage. They have, nevertheless, made it into clinical use. **Methods:** This paper presents a literature assessment of commercial medical systems with autonomous and semi-autonomous features, as well as experimental work involving the automation of several surgical operations. The results are gathered from significant databases, with the exception of articles that have not been tested on real robots. **Results:** Our search turned up a number of experimental and clinical papers describing advancements in autonomous surgical manoeuvres, ultrasound guidance, optical coherence tomography guidance, cochlear implantation, motion compensation, orthopaedic, neurological, and radiosurgery robots, among other topics. **Conclusion:** Autonomous and semi-autonomous systems are starting to appear in a variety of treatments, automating key phases in the process. These devices are likely to become standard practise and transform the surgical landscape.

Medical Robot Manipulator for Ultrasound Scanning

In normal clinical procedures, ultrasound (US) capture necessitates manual probe operation based on picture interpretation. A robotic system for autonomous US acquisitions has a lot of potential to reduce sonographers' burden, enhance access to treatment, produce more uniform imaging findings, and avoid direct patient interaction. They have presented an overview of the current state of autonomous robotic US acquisition systems and approaches in the literature in this study [4] (Li, Xu and Meng, 2021). In particular, they propose a framework for standardising increasing degrees of autonomy in robot ultrasonography, as well as a brief discussion of non-autonomous and semi-autonomous systems. Then, based on the hardware components, process, and applications, they have analysed the present autonomous robotic systems designed for extracorporeal US acquisitions. In addition, they have review three key aspects connected to autonomous US acquisitions: scanning route planning, contact force control, and picture quality enhancement, and discuss current methodologies and strategies offered on each topic shown in figure 3. Finally, they have analysed the obstacles and suggest some prospective research avenues. They think that the current efforts in autonomous robotic acquisitions in the United States will pave the way for a bright future in medical care and help a wide range of clinical operations.

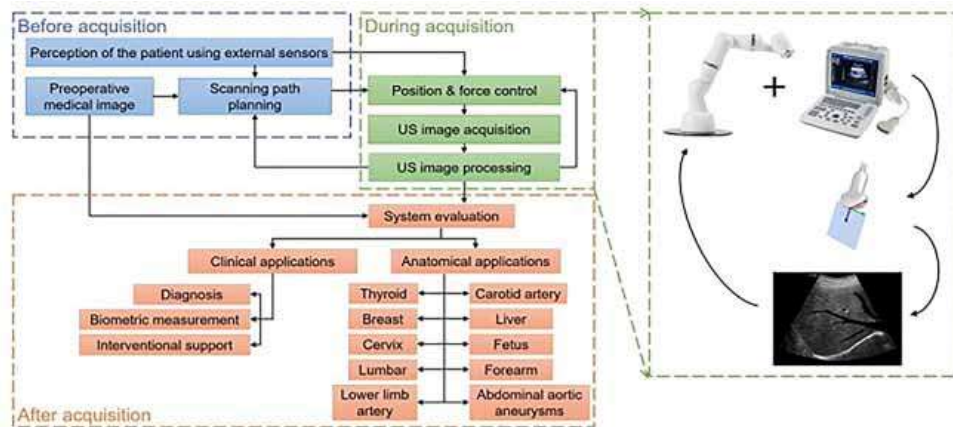


Figure 3

The goal of this study [5] (von Haxthausen et al., 2021) is to give an overview of the most recent robotic ultrasound systems that have appeared in the last five years, as well as their current status and future directions. The systems are classified according to their degree of robot autonomy (LORA). From recent findings, this review states that teleoperating systems are the most technologically advanced. The research phase for collaborative assistance and autonomous systems is currently underway, with an emphasis on ultrasonic image processing and force adaption techniques. Clinical investigations and proper safety methods, however, are missing essential components. To increase picture understanding and ergonomics, future research will likely rely on artificial intelligence and virtual/augmented reality. A review of robotic ultrasound systems is provided, along with the initial technical requirements shown in figure 4. The literature from the last five years is classified into three categories: teleoperation, collaborative assistance, and autonomous systems based on LORA. Finally, future prospects in robotic ultrasound systems are discussed, with artificial intelligence and virtual/augmented reality being highlighted.

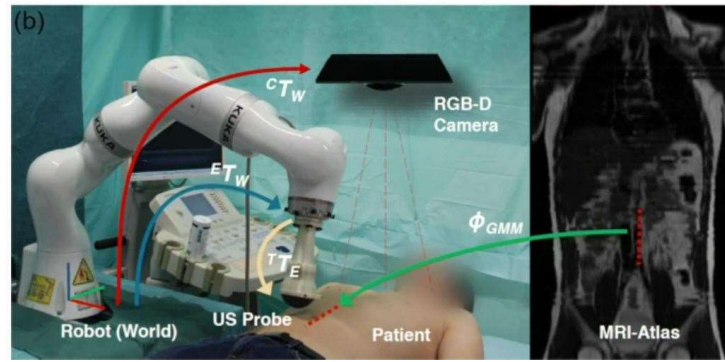


Figure 4

According on expert demonstrations [6] (Mylonas et al., 2013), they offer a learning-based controller to enable autonomous execution of the eFAST scanning by a lightweight robotic manipulator. This strategy has two distinct advantages. First, without requiring complicated robotic teleoperation, the automatically obtained USS pictures may be delivered to a professional radiologist from a remote site. Second, using learning by demonstration reduces the complexity of robotic programming and enables for the natural and intuitive extraction of operator-specific information IN SITU. The addition of force information can increase the system's adaptability by allowing for quick adaption to varied dynamic contexts.

Medical ultrasonography (US) devices are commonly employed for interior tissue diagnostics. However, there are difficulties in collecting and interpreting US pictures, such as poor probe placement and a lack of spatial information. In this study [7] (Suligoj, Heunis, Sikorski and Misra, 2021), they have used a robotic framework with a high level of autonomy to extend the possibilities of medical US imaging. A 3D camera captures the surface of a humanoid phantom as a point cloud, which is subsequently utilised to design the US probe's course and navigation. An impedance controller maintains steady contact with the surface during US scanning and adjusts for uneven and shifting surfaces, allowing for robotic probe placement. The placement accuracy of robots in the United States is 1.190.766mm. On static surfaces, the mean force along the z-direction of the US probe is 6.111.118N, whereas on moving surfaces, it is 6.632.18N. During scanning, the lowest observed force of 1.58N confirms consistent probe-to-surface contact. The surface and interior anatomical features of the phantom are 3D reconstructed and multi-modally seen using acquired US images. Finally, distinct tissues are segmented using K-means clustering. According to the Jaccard similarity coefficient, the best segmentation accuracy of the jugular vein is 0.89. With such precision, this approach might significantly increase autonomous US collection and clinician diagnostic confidence.

Because ultrasonic scans are safe, affordable, and non-invasive, physicians utilise them to diagnose by creating real-time pictures of inside organs. However, due to an ageing populace and a shortage of physicians, diagnosing patients in rural locations is challenging. Furthermore, the technical abilities of the physician have an impact on the interpretation of the scans and consequently the examination outcomes. To address the issues, the creation of a robotic system for remote diagnostics is necessary. They have proposed the creation of a robotic system for automated ultrasound imaging of the human liver for this aim [8] (Mustafa et al., 2014). Several elements created for the robotic system are presented in this study shown in figure 5. The first component is an algorithm for predicting the location of the liver in order to direct the probe to the epigastric area for scanning. The construction of an ultrasonic probe scanning routine to

acquire a complete picture of the liver is the second part. The third component is a force control algorithm that keeps the probe in touch with the human body even when the patient is breathing. When erroneous ultrasound pictures appear, the fourth element is an algorithm for identifying and correcting the probe's location. These algorithms, along with an implementation software, operate the equipment, which was tested on human subjects and is a Mitsubishi Electric MELFA RV-1 six-axis manipulator. The outcomes validated the strategy's success.

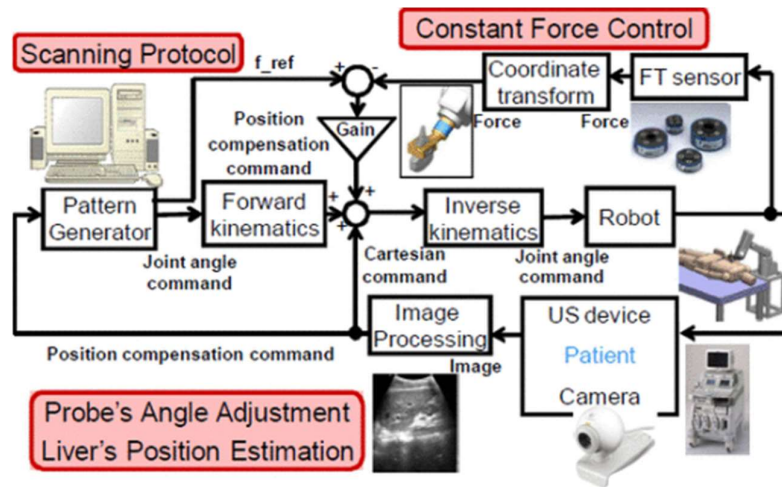


Figure 5

Force, control and accuracy of Robot Manipulator

The control design of medical robot ultrasound for correctly scanning the liver has a big difficulty in compensating for respiratory motion. In order to enhance the guide performance for anatomical or pathological features of the liver during free breathing, this work provides an adaptive fuzzy proportional–integral–derivative (PID) force control of robot-assisted ultrasonography. The liver guiding process is recreated using a six-degree-of-freedom robotic arm with a 2-D ultrasound probe and a force sensor. Retrospective hepatic ultrasound scans are also used to simulate and coordinate respiratory movements. The developed force controller uses fuzzy logic to directly auto-tune the PID controller gains for manipulating the robotic ultrasound probe on the patient's abdomen at desired force levels in real time, without requiring prior knowledge of the robot's nonlinear dynamic characteristics or mathematical modelling. Using four ultrasound imaging sequences of the liver, a simulation framework was constructed to evaluate the robotic force controller. The adaptive fuzzy PID controller outperformed the traditional PID controller with fixed gains, resulting in minimal tracked force errors of around 0.3 and 0.01 N with and without force sensor noise, respectively, for all picture datasets examined. Based on the established fuzzy PID controller, this simulation research [9] (Karar, 2017) might be a useful way to fulfil the tracking task of required probe forces during robotic liver ultrasound guiding.

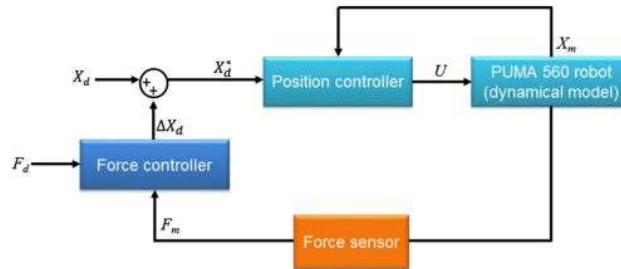


Figure 6

In today's clinical evaluations, ultrasound scanning is crucial. It has been widely utilised in yearly physical examinations and many other diagnosis and intervention procedures because to its tiny footprint, low cost, and popularity. The scanning findings, on the other hand, are highly dependent on the clinician operators' abilities, resulting in inaccuracy and even false detection. A completely autonomous ultrasonic scanning robot might be a potential option, allowing more precise and consistent operation. In this study [10] (Huang et al., 2021), they have offered a procedure-specific imitation learning framework for autonomous scanning tasks based on clinical protocols. They have built a feature-based visual servo controller for both in-plane and out-of-plane feature tasks based on the targeted carotid artery inspection process. A one-step-exploring (OSE) strategy improves the resilience and smoothness of transitions between jobs. Experiments using medical phantoms and human participants are used to test and validate the proposed approaches. The results demonstrate that the suggested approach may increase the procedure completion success rate from 22.2 percent to 84.6 percent for out-of-plane feature-based operations.

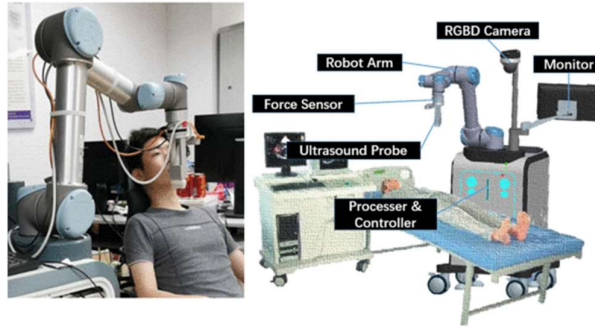


Figure 7

Purpose In complicated soft unpredictable clinical contexts, autonomous ultrasound imaging via robotic ultrasound scanning devices is crucial and hard to aid in therapy. They have presented an autonomous robotic ultrasound (US) control approach based on reinforcement learning (RL) model to create the link between the environment and the system to cope with the complicated environment experienced by the ultrasound probe during the scanning process. Without any a priority information about the target or the surroundings, the suggested approach requires just contact force as input information to accomplish robot control of the probe's posture and contact force. Methods to understand the link between contact force and output force directly, a policy gradient theorem-based RL model is presented and trained with the 6-degree-offreedom (DOF) contact force of the US probe. Then, using the required contact force as the action space, a force control strategy based on the admittance controller is given for synchronous force, orientation, and position management. By scanning an unknown soft phantom, the suggested approach was assessed using acquired US pictures, contact force, and scan trajectories has provided some results. In the US photos, the suggested method differs from the free-hand scanned approach by 30.4 percent, according to the testing data. Their approach was able to create stable scanning processes on a soft uncertain skin surface and acquire US pictures, according to the analysis findings of contact forces and trajectories. Based on reinforcement learning, they have concluded a brief and efficient force-guided US robot scanning control approach for soft uncertain environments. The volunteer experiments indicated the potential application value of our method in the complex clinical environment of robotic US imaging system, especially with limited visual information, and the experimental results validated our method's feasibility and validity for complex skin surface scanning [11] (Ning, Chen, Zhang and Liao, 2021).

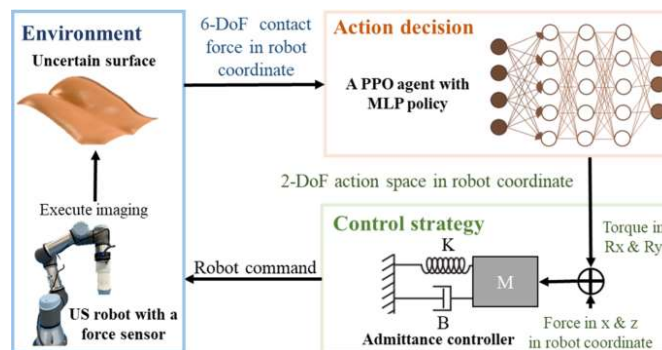


Figure 8

Medical ultrasonography is widely used to identify tissue textures and characterise lesions, and it is the preferred method for thyroid disease identification and monitoring. An occupational operator uses a hand-held ultrasound probe to conduct traditional medical ultrasound operations. These procedures are physically and mentally taxing, and the clinical outcomes are strongly dependent on the operator, eroding faith in ultrasound imaging data accuracy in routine assessments. In contrast, a robotic ultrasound technique is a new paradigm that combines a robotic arm with an ultrasound probe. Controlling the scanning trajectory, region of interest, and contact force allows for automated or semi-automatic ultrasonic scanning. As a result, scanning becomes more informative and comparable over time in successive tests. They have offered a system that uses a mix of predetermined trajectory execution and real-time force feedback control to allow operators to successfully replicate similar ultrasound pictures. A 7-axis robotic arm with 6-DoF force-torque sensors and a linear-array ultrasound probe are included in the platform. During autonomously performed exams, the measured forces and torques impacting the probe are used to adaptively adjust the predetermined trajectory, and probe-phantom contact force accuracy is assessed. In parallel, structural characteristics may be recovered from the scanned volume through a 3D scan by processing and merging ultrasonic B-Mode pictures with probe spatial information. They did the validation on a tissue-mimicking phantom with thyroid characteristics, and they were able to exhibit good image registration accuracy over several trials [12].

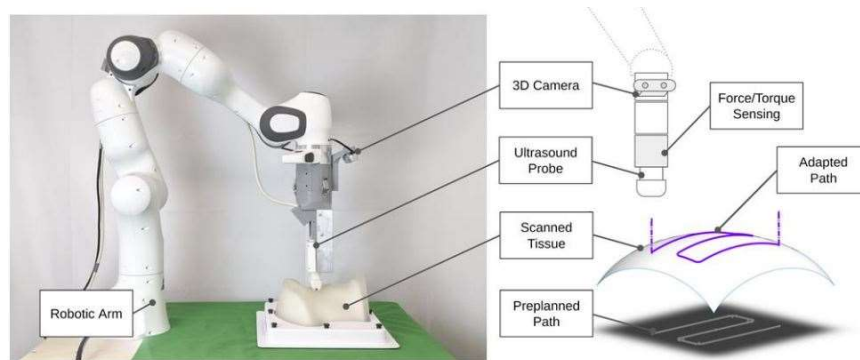


Figure 9

Localization using computer vision and Landmarks properties

Scoliosis development in teenagers need constant monitoring in order to adopt therapeutic steps in a timely manner. In scoliosis assessment, ultrasound imaging offers a non-radioactive alternative to X-ray, which is commonly utilised in clinical practise. However, speckle noise in ultrasound pictures makes it difficult for sonographers to discern bone structures and trace the curvature of the spine. For scoliosis evaluation, this work [13] (Victorova, Lee, Navarro-Alarcon and Zheng, 2022) presents a unique robotic ultrasound technique for spinous process localisation and automated spinal curvature tracking. A fully linked network with a deconvolutional head is used to calculate the locations of the spinous processes. A collection of ultrasound pictures from 25 human participants with scoliosis was subjected to a 5-fold cross-validation. With a mean distance error of 1.0 0.99mm, the proportion of accurate key points of the spinous process is 0.966 0.027. They employed their machine learning-based technology to steer the mobility of the robot-held ultrasound probe while recording ultrasound pictures and to monitor the spinal curvature. They offered a novel force-driven controller that automatically adapts the probe's posture and orientation to the skin surface, ensuring optimal acoustic coupling between the probe and the skin. They have employed the network architecture to classify the spine into areas, such as sacrum, lumbar, and thoracic, which are used to alter the probe's orientation to accommodate for the variable curvature along the spine. Following the autonomous scanning, the gathered data is utilised to create a coronal spinal picture, from which the scoliosis spine deformity may be diagnosed and measured. Traditional approaches based on phase symmetry and picture intensity were compared to the suggested learning-based method for anatomical landmarks localisation. The learning-based technique found to be more exact in localising spinous processes while processing pictures at a quicker pace, which is beneficial for real-time scoliosis scanning. They conducted an experimental investigation using human scoliosis individuals to test the performance of our robotic technique, comparing deviations of the spinous process from the picture centre to those seen in a manual scan. Their findings demonstrate that the robotic technique decreases mean spinal curvature error from 4.6 4.6mm (manual scanning) to 1.0 0.8mm (robotic scanning) for mild scoliosis and from 4.3 3.9mm (manual scanning) to 2.8 1.8mm (robotic scanning) for moderate scoliosis. Both approaches have equivalent angles of spinal deformity measured on spinal reconstruction pictures, showing that they equally mirror human anatomy. The scanning performance was improved by using a spinal region-specific moment-based probe orientation control. To determine the relevance of each component of the proposed system, ablation research was conducted.

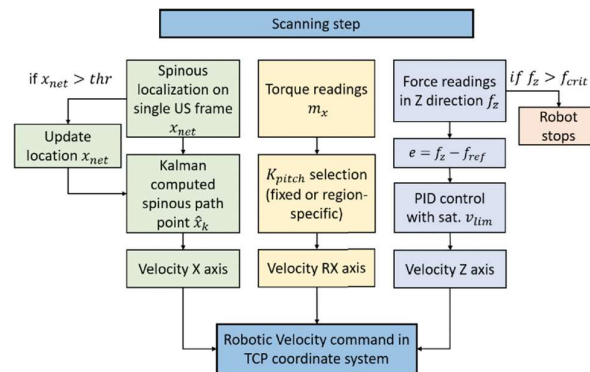


Figure 10

Ultrasound scans are used by doctors to acquire real-time pictures of interior organs since they are safe and affordable. Due to an ageing culture and a physician scarcity, persons in distant places may find it difficult to be scanned. As a result, developing an autonomous robotic system to do distant ultrasound exams is critical. They had previously built a robotic device for doing an automated ultrasound scan of the human liver. They have offered in this study [14] a method for autonomously localising the epigastric area as the beginning place for the automated ultrasonic scan in order to make it a totally autonomous system. On a digital shot of the patient's belly, an image processing programme highlights the umbilicus and mammary papillae. Then, considering the distances between these landmarks, we estimated the position of the epigastric area. Using the connection between force and displacement, a supporting algorithm identifies rib position from epigastrium. These algorithms were used with the automated scanning system of a Mitsubishi Electric MELFA RV-1 six-axis manipulator. The device correctly found the epigastric area 94% of the time in tests on 14 healthy male volunteers. The findings indicate that picture recognition was successful in locating a human body component.

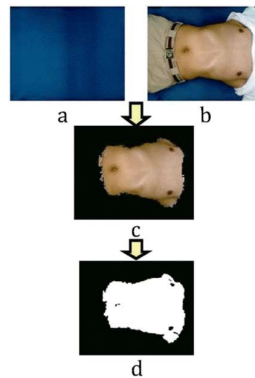


Figure 11

Medical imaging has ushered in the largest paradigm shift in contemporary medicine's history, with ultrasound (US) being the most widely used imaging modality. The incorporation of digital imaging into the surgical arena offers up new diagnostic and interventional possibilities, while the use of robots improves precision and targeting capabilities. This study [15] (Elek et al., 2017) examines the present state of robotic platforms in the United States, outlining key research and clinical trends as well as existing capabilities and constraints. The study focuses on non-autonomous US-based systems, automated robotic navigation systems located in the US, and guided autonomous tools based in the US. These domains chart the course for future research, predicting a slew of new applications in the field of computer-assisted surgery.

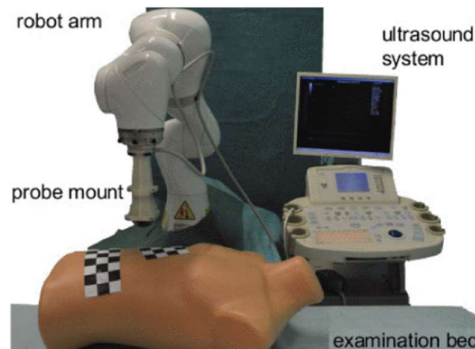


Figure 12

In computer aided surgery, surgical navigation is critical. The employment of robots in the operating room has benefited greatly from recent breakthroughs in this idea, particularly with the use of non-invasive and radiation-free imaging methods such as ultrasonography (US). The methods and software tools created for orthopaedic surgical navigation in the United States are presented in this study [16] (Gonçalves et al., 2014). The techniques and software tools created for operating a robotic manipulator are provided in the second half of the study, which may be utilised both in simulation and in a real-world scenario. The generated simulation scenario for a surgical operation in Hip Resurfacing, such as the initial drill of the femur head, is also shown in the study. During surgery, the system acquires a 3D US bone surface from a series of US pictures for guidance. This bone surface will then be registered to the pre-operative bone model for a more exact understanding of bone position and orientation. There are two phases to this registration. The first is a global registration prior to surgery to precisely register the bone. The second option is to record the femur locally, which is quicker and better for tracking bone motions. The robot manipulator updates its drilling position and orientation based on the measured bone movement.

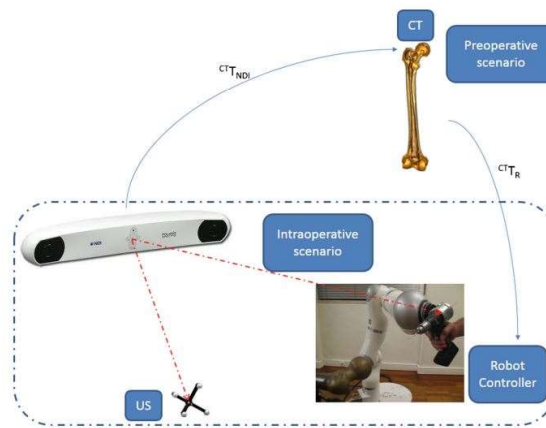


Figure 13

Ethics and safety Precaution for Medical Robots

Medical robotics is one of the fastest expanding industries in the medical technology business, with applications ranging from minimally invasive surgery, targeted therapy, and hospital optimization to emergency response, prostheses, and home support. The regulatory, ethical, and legal constraints that medical robots face need careful consideration of various levels of autonomy as well as the environment in which they will be used. Levels of automation for on-road vehicles are established for autonomous cars, but no similar specifications exist for medical robots. They suggested six levels of autonomy for medical robots as one possible framework to spark debate shown below [17] (Yang et al., 2017).



Figure 14

The goal of this paper [18] (O'Sullivan et al., 2019) is to advance the conversation about the potential of artificial intelligence (AI) and autonomous robotic surgery, with a particular focus on ethics, regulation, and legal issues (such as civil law, international law, tort law, liability, medical malpractice, privacy, and product/device legislation, among other things). They researched existing and developing AI and autonomous technologies (e.g., cars), military and medical technologies (e.g., surgical robots), applicable frameworks and standards, cyber security/safety, and legal systems throughout the world. They discussed the special issues that ideas for AI in general (e.g., Explainable AI) and machine learning in particular (e.g., black box) provide for robotic surgery, as well as recommendations for building and upgrading applicable frameworks or standards. They divided responsibility into three categories: accountability, liability, and culpability. Whether these were civilian or military patients, these three factors were discussed when addressing accountability for AI and autonomous surgical robots (however, these aspects may require revision in cases where robots become citizens). Culpability is the component that offers the least clarity, as it is unimaginable in today's technological condition. They envisioned a surgical robot that could learn and execute common operating duties while being overseen by a human surgeon in the near future. This is a surgical analogy to self-driving cars. As a 'doctor in the loop,' a person stays in the 'driving seat,' safeguarding patients undergoing procedures backed by surgical robots with autonomous capabilities.

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